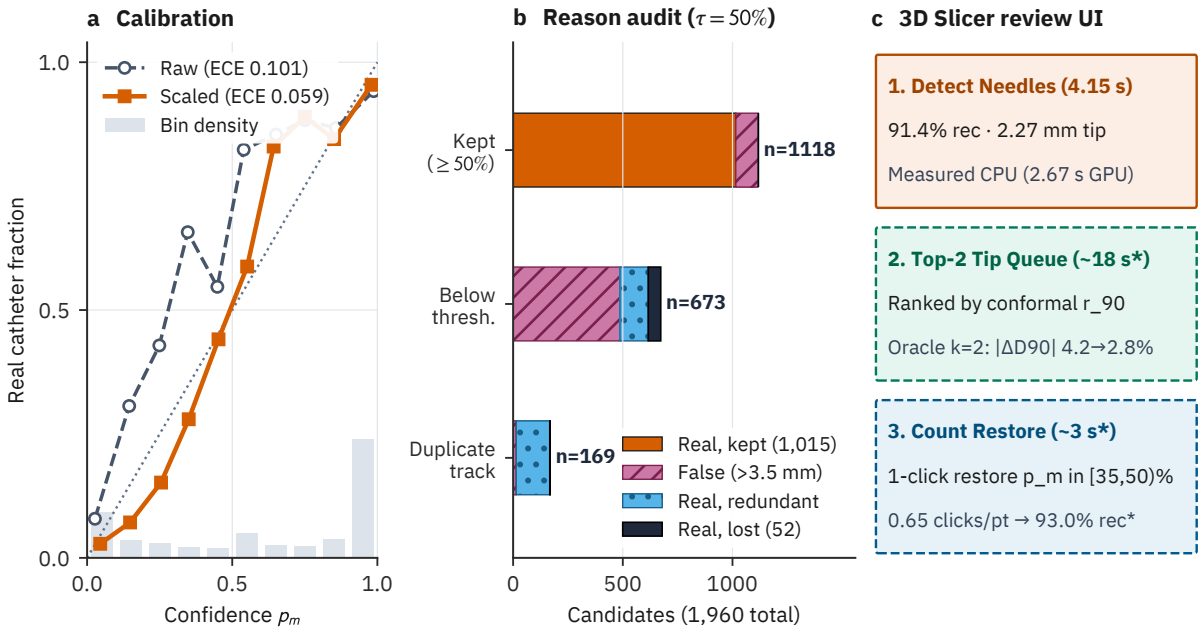
a Calibration
Raw (ECE 0.101)
Scaled (ECE 0.059)
Bin density
Real catheter fraction
1.0
0.5
0.0
0.0
0.5
1.0
Confidence p_m
b Reason audit ($\tau = 50\%$)
Kept (≥ 50%)
Below thresh.
Duplicate track
n=1118
n=673
n=169
Real, kept (1,015)
False (>3.5 mm)
Real, redundant
Real, lost (52)
0
500
1000
Candidates (1,960 total)
c 3D Slicer review UI
1. Detect Needles (4.15 s)
91.4% rec · 2.27 mm tip
Measured CPU (2.67 s GPU)
2. Top-2 Tip Queue (~18 s*)
Ranked by conformal r_90
Oracle k=2: |ΔD90| 4.2→2.8%
3. Count Restore (~3 s*)
1-click restore p_m in [35,50]%
0.65 clicks/pt → 93.0% rec*